

PUMP MASTER PRO™

Next-Generation Industrial Pump Sizing, CAD Pipe Network Modeling & Submittal Automation

[✓] ISO 9906:2012 (Grade 1 & 2) [✓] ANSI/HI 14.6 Test Methods [✓] API 610 (11th/12th Edition) [✓] Pipe Network Solver

“From Raw Operating Duty Point to Publication-Grade Customer Submittal in Under 3 Minutes.”

Pump Master Pro is the cloud-native hydraulic sizing, curve-fitting, and pipe network analysis platform built specifically for Centrifugal Pump Manufacturers (OEMs), EPC Contractors, Consulting Engineers, and Heavy Industrial Plants (Mining, Water & Slurry). By replacing disjointed spreadsheets and manual curve interpolation, Pump Master Pro unifies duty point selection, friction loss solving, viscous/slurry deratings, and publication-ready PDF datasheets into a single collaborative suite.

1. Target Audiences & Value Delivered

Industry / Stakeholder	Traditional Bottleneck	Pump Master Pro Solution
Pump OEMs & Sales Distributors	Slow quotation turnaround (2–4 hours per quote) and risk of sales reps recommending off-design pumps.	Instant duty point matching, automated impeller trims, branded submittal generation, and multi-tenant catalog hosting.
EPC & Consulting Engineers	Error-prone spreadsheet friction formulas and tedious manual NPSHa safety margin checks.	Interactive orthogonal CAD pipe network modeling, automated Colebrook-White friction solving, and system curve overlay.
Mining & Slurry Plants	Unpredicted pipe sanding, line blockage, or catastrophic pump wear due to incorrect slurry settling physics.	Native Durand critical deposition velocity (Vc) and Ferguson & Church particle settling velocity algorithms.
Water & Wastewater Utilities	Operating off Best Efficiency Point (BEP), uncontrolled energy waste, and cavitation damage in parallel pumps.	Comprehensive POR/AOR envelope visualization, VFD speed affinity curves, and multi-pump side-by-side comparison.

2. The 6 Core Engineering Modules

■ Module 1: Intelligent Pump Sizing & BEP Analysis

- Multi-Parameter Search: Filters thousands of pump curves instantly by Flow (Q), Head (H), and available NPSH.
- BEP Proximity & Reliability: Highlights Preferred Operating Regions (POR: 70–120% of BEP) and Allowable Operating Regions (AOR) to prevent premature bearing, seal, and impeller failure.
- Affinity Law Scaling: Automated impeller trim diameter calculations (D/D_{max}) with Karassik efficiency deratings, plus Variable Speed Drive (VFD / RPM) operating curves.
- Multi-Pump Comparison: Compare up to 4 pumps side-by-side on superimposed H-Q, Efficiency, Power, and NPSHr charts.

■ Module 2: Interactive CAD Pipe Network Designer

- Orthogonal CAD Routing: Clean horizontal and vertical pipe routing with smooth rounded integrated elbows, reducers, tees, and inline valves.
- Exact Friction Physics: Solves Darcy-Weisbach head loss using iterative Colebrook-White friction factor algorithms—eliminating the severe fluid limitations of Hazen-Williams.
- System Curve Generation: Automatically computes static head (Δz) plus dynamic friction loss ($k \cdot Q^2$) and overlays the system curve directly onto pump performance curves.
- Fittings Database: Built-in equivalent length and K-factor library for elbows, check valves, foot valves, gate valves, and strainers.

■ Module 3: Slurry & High-Density Solids Engine

- Critical Deposition Velocity (Durand Method): Calculates V_c to guarantee operating velocity prevents solids dropout and pipeline sanding.
- Particle Settling Velocity: Computes Ferguson & Church settling velocity (V_t) based on particle size distribution (d50 to d90) and carrier liquid properties.
- Slurry Derating Ratios: Automatically determines Head Ratio (HR) and Efficiency Ratio (ER) to compute real-world slurry shaft brake power (kW).

■ Module 4: Viscous Fluid Modeling (ANSI/HI Standard)

- Hydraulic Institute Correction: Native implementation of ANSI/HI 9.6.7 procedures for viscous liquids up to thousands of centistokes (cSt).
- Dynamic Derating Factors: Real-time calculation of CH (head derating), CQ (flow derating), and C_η (efficiency derating) as users enter viscosity and operating temperature.

■ Module 5: High-Precision Curve Digitization & Fitting

- Automated Polynomial Regression: Fits discrete manufacturer test points into high-order smooth polynomial curves with $R^2 > 0.998$.
- Complete Curve Quad-Stack: Full simultaneous generation of H-Q, Hydraulic Efficiency (η), Shaft Brake Power (kW), and NPSH Required curves.
- Water Vapor Pressure Solver: Built-in Antoine / saturation vapor pressure modeling (P_v) accounting for elevation altitude and operating water temperature.

■ Module 6: Publication-Grade Technical Submittals & PDF Reports

- Instant Commercial Submittals: Generates clean, publication-grade A4 engineering datasheets ready for client tenders and project submittals.
- Customizable Visual Layouts: Flexible 1, 2, 3, or 4 stacked performance charts with configurable curve line styles, colors, thickness, and grid patterns.

- Corporate White-Labeling: Seamless injection of manufacturer/contractor logos, contact headers, project names, and engineering approval remarks.

3. Business Impact & Return on Investment (ROI)

Key Performance Metric	Traditional Method (Excel / Desktop)	With Pump Master Pro	Net Impact / ROI
Quotation Turnaround Time	2 to 4 hours per complex pump quotation.	Under 3 minutes end-to-end.	90% Time Savings
Selection & Calculation Accuracy	Manual formulas prone to broken cells and unit conversion errors.	Automated server-side Python engineering solvers.	Zero Math Errors
Tender Win Rate	Static text tables and poor-quality scans/screenshots.	Professional branded datasheets with vector CAD curves.	+35% Higher Bid Success
Field Cavitation & Reliability	Improper NPSHa safety margins leading to early impeller wear.	Explicit POR/AOR checks and NPSH margin validation.	Fewer Warranty Claims

4. Architecture, Security & Deployment

- 100% Cloud-Native SaaS: Zero desktop installations, DLL dependencies, or license dongles. Works seamlessly in any modern browser on Windows, Mac, Linux, and tablets.
- Server-Side Intellectual Property Vault: All proprietary equations (Darcy-Weisbach, Ferguson & Church settling, Durand velocity, Karassik trimming, and vapor pressure) execute strictly inside a hardened server-side calculation engine. Browsers never receive raw formulas.
- Organization & Multi-Tenant Management: Role-based permissions allow corporate administrators to manage teams, distribute catalogs, and govern permissions between Lead Engineers, Sales Reps, and Clients.

Ready to Modernize Your Pump Sizing & Quotation Workflow?

Experience automated pump curves, CAD pipe network analysis, and instant engineering submittals today.

🌐 Website: <https://pumpmasterpro.com> | ✉ Contact & Access: admin@pumpmasterpro.com